MIL-C-55172B(EL)
17 January 1972

SUPERSEDING
MIL-C-55172A(EL)
21 September 1965

MILITARY SPECIFICATION

CABINET, ELECTRICAL EQUIPMENT, CY-3397( )/G

1. SCOPE

1.1 Scope.- This specification covers one type of electrical equipment cabinet to house communications equipment requiring standard 19 inch rack mounting designated as Cabinet, Electrical Equipment, CY-3397( )/G. (See 6.3).

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder, Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
QQ-S-781	Strapping, Flat; Steel.
PPP-B-20	Bags, Cotton, Mailing.
PPP-B-585	Box - Wood, Wire Bound.
PPP-B-601	Boxes; Wood, Cleated - Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-T-45	Tape, Gummed, Paper, Reinforced.
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

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MIL-P-116	Preservation, Methods of.
MIL-W-6858	Welding, Aluminum, Magnesium, Non-hardening Steels or Alloys, and Titanium, Spot, Seam, and Stitch.
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric acid base).
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-B-43014	Boxes, Water Resistant Paperboard; Folding, Set-up, and Metal-Stayed.

STANDARDS

MILITARY

MIL-STD-22	Welded-Joint Designs.
MIL-STD-105	Sample Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment Classification of Visual and Mechanical Defects.
MIL-STD-454	Standard General Requirements for Electronic Equipment.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-433898	Cabinet, Electrical Equipment, CY-3397( )/G.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description.- Cabinet, Electrical Equipment, CY-3397( )/G is a general purpose cabinet for housing communications equipment requiring standard 19 inch rack mounting.

3.2 Construction.- The equipment shall be constructed in accordance with the requirements of this specification and SC-DL-433898.

3.3 Cleaning.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

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3.4 Finish, protective.- Equipment shall be given protective finish in accordance with MIL-F-14072 and the equipment drawings. (See 4.3).

3.5 Marking.- Marking shall conform to MIL-M-13231. (See 4.3).

3.6 Welding.- Wherever practicable, welded joints shall conform to MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall be Class A in accordance with MIL-W-6858. When spot or other intermittent welds are used to hold a part, the number of welds shall be at least two.

3.6.1 Cleaning prior to welding.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.6.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved.

3.6.3 Cleaning after welding.- Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with MIL-M-10578. After cleaning, acid used shall be completely neutralized and removed.

3.7 Soldering.- Securely solder all electrical connections, except the crimp-on and screw connections, using solder in accordance with QQ-S-571, Composition SN60, Type RMA.

3.8 Wiring.- Wiring shall have sufficient slack to prevent undue stress and to permit at least two replacements of the part to which the wire is connected. Splicing of conductors is not permitted. Wiring shall be properly supported and not depend on soldered connections for mechanical support.

3.9 Moisture resistance.- The equipment shall meet the requirements of 3.10 and 3.11, and shall show no evidence of deterioration after being subjected to the test specified in 4.9.

3.10 Continuity.- When tested as specified in 4.7, each conductor shall be electrically continuous.

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3.11 Breakdown voltage, insulation.- There shall be no evidence of a voltage drop when tested in accordance with 4.8.

3.12 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454. (See 4.5).

3.13 Workmanship.- The equipment shall be manufactured and assembled in accordance with applicable portions of the following paragraphs herein: (See 4.6).

3.3 Cleaning	3.6 Welding
3.4 Finish	3.7 Soldering
3.5 Marking	3.8 Wiring

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

4.2.1 Inspection covered by subsidiary documents.- (See 4.3).

4.2.2 Quality conformance inspection.-

(a) Quality conformance inspection of equipment before preparation for delivery. (See 4.4).

(b) Quality conformance inspection of preparation for delivery. (See 4.10).

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4.3 Inspection covered by subsidiary document.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.5

4.4 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.3 and 4.4.1 through 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specifications requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.4.1 Group A inspection.- Each unit (100%) of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of table I. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQLs indicated in table I. Group A inspection shall be performed in any order which is satisfactory to the Government.

Table I - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL	
			Major	Minor
Visual and mechanical	3.13	4.6	1.0%	4.0%
Electrical				
Continuity	3.10	4.7	1.0% combined	
Breakdown voltage	3.11	4.8		

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4.4.2 Group B inspection.- This inspection, including sampling, shall conform to table II and to special procedures for small sample inspection of MIL-STD-105. The inspection level shall be S-4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The AQL shall be as shown in table II.

Table II - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Interchangeability	3.12	4.5	6.5%

4.4.3 Group C inspection.- This inspection shall consist of the tests specified in table III, and shall be performed on one unit that has been subjected to and met group A and group B inspection. Sample unit shall be selected at random without regard to quality.

Table III - Group C inspection

Inspection	Req. Para.	Insp. Para.
Moisture resistance	3.9	4.9

4.4.3.1 Noncompliance.- The contractor shall immediately report, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed at the option of the Government) and group A and group B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the group C inspection results have shown that the corrective action was effective.

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4.4.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection may be accepted on contract, provided they are resubjected to and pass group A inspection after repair of all visible damage.

4.5 Inspection for dimensional interchangeability.- The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.12. When a listed dimension is not within specified limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

4.6 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in MIL-STD-252 and table IV. (See 3.13).

Table IV - Classification of visual and mechanical defects

Classification	Defect
Major	1. Electrical receptacle missing or broken. 2. Switch inoperative or missing. 3. Wire broken. 4. Insulation cut or torn. 5. Hinge missing. 6. Improper fit of door. 7. Welds missing. 8. Sub-standard welds. 9. Scratches, cuts, abrasions, etc. with exposure of bare metal.
Minor	1. Loose screws, nuts and electrical accessories. 2. Lock inoperative. 3. Scratches, cuts, abrasions, etc. with no exposure of bare metal. 4. Fuse missing. 5. Burrs and sharp edges. 6. Color not as specified.

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4.7 Continuity test.- Electrical continuity of wiring and other connections, including ground connections, shall be verified by a point-to-point test. (See 3.10).

4.8 Breakdown voltage insulation test.- A potential of 600 volts rms shall be applied instantaneously and maintained for ten (10) seconds between each electrical circuit and cabinet ground. Before testing, disconnections shall be made where necessary to protect circuit elements. Switches shall be in the ON position.

4.9 Moisture resistance test.- (See 3.9).

4.9.1 Test method.-

(a) Dry at 130° F ± 5° for 24 hours.

(b) Condition at 77° F ± 5° and 40 to 50 percent relative humidity for 24 hours.

(c) Perform the tests specified in 4.7 and 4.8.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and a period of time for each portion of the cycle shall conform to MIL-STD-170. No electrical measurements shall be made during cycling.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° F ± 5° and 40 to 60 percent relative humidity. The equipment shall meet the requirement specified in 3.9.

4.10 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C as specified. (See 6.2).

5.1.1 Level A.-

5.1.1.1 Cleaning.- Cabinet, Electrical Equipment CY-3397( )/G shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying.- Cabinet, Electrical Equipment CY-3397( )/G shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Unit packaging.- Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Screws.- Place the screws in a bag, conforming to PPP-B-20. Secure the bag with the drawstrings.

5.1.1.4.2 Touch-up paint and paint brush.- Each jar of touch-up paint and each paint brush shall be packaged individually method III as follows: Wrap each item in paperboard conforming to PPP-P-291, type I, style 2. Secure wraps with tape conforming to PPP-T-45, type III, grade A.

5.1.1.4.3 Consolidated package.- Consolidate the items, packaged as specified in 5.1.1.4.1 and 5.1.1.4.2, within a close-fitting paperboard box conforming to MIL-B-43014, Form 1, style II, type A, class a. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.4 Cabinet, Electrical Equipment CY-3397( )/G.- Each Cabinet, Electrical Equipment CY-3397( )/G shall be packaged method III as follows: Place the consolidated package, packaged as specified in 5.1.1.4.3, within the cabinet. The consolidated package shall be adequately secured to the interior of the cabinet with tape conforming to PPP-T-76. Cushion the cabinet on all surfaces with cells and pads or both fabricated of fiberboard conforming to PPP-F-320, Type CF, Class Weather-resistant, Variety SW, Grade V3c. Place the cushioned cabinet within a close-fitting box conforming to PPP-B-636, V3c. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C.- Cabinet, Electrical Equipment CY-3397( )/G shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling, and limited in transit storage.

5.2 Packing.- Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2).

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5.2.1 Level A.- Cabinet, Electrical Equipment CY-3397( )/G, packaged as specified in 5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. Fiberboard boxes being placed directly in the shipping container shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. Skids, 3 x 4 inches, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.- Cabinet, Electrical Equipment CY-3397( )/G, packaged as specified in 5.1, shall be individually packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. Skids, 3 x 4 inches, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C.- Cabinet, Electrical Equipment CY-3397( )/G, packaged as specified in 5.1, shall be packed in shipping containers that comply with Federal or Military specifications, in a manner that will afford adequate protection to the package and its contents against physical and environmental deterioration and damage, during limited shipment, handling and in transit storage.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The cabinet covered by this specification is intended to house communication equipment requiring standard 19 inch rack mounting.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of preservation, packaging, and packing. (See Section 5).

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) When rough handling and functional tests are required.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: CY-3397W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.4 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-55172B(EL) CABINET, ELECTRICAL EQUIPMENT, CY-3397( )/G

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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